Anfaz Edakkavil Aboobakar

Software Engineer | AI & Data

+353 892007825 | anfaze98@gmail.com | linkedin.com/in/anfaz-e | github.com/ft-anfaaz | portfolio-anfaz.vercel.app |
Cork, Ireland (Open to relocate) | Stamp 2

Profile

Software Engineer with nearly 4 years of experience developing and supporting enterprise insurance solutions using XQuery, Java, SQL, and RESTful APIs. Skilled in document-generation systems, query optimisation, production deployments, Agile delivery, and cross-functional collaboration. MSc graduate in Artificial Intelligence with hands-on experience in machine learning, deep learning, natural language processing, big data processing, transformer models, multimodal AI, and explainable AI.

Education

Master of Science in Artificial Intelligence (78.6%) | September 2025 – September 2026

Munster Technological University, Cork, Ireland

Module: Practical Machine Learning, Deep Learning, Natural Language Processing, Big Data Processing, Knowledge Representation and Reasoning, Decision Analytics, Metaheuristic Optimisation, and Software Agility

Bachelor of Engineering in Computer Science (7.74 CGPA) | July 2017 – June 2021

Sahyadri College of Engineering and Management, Mangalore, India

Module: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Software Testing, Web Technologies, and Python Programming

Experience

Technology Analyst | August 2021 – July 2025

Infosys Limited, Bangalore, India

- Promoted three times over four years, from Systems Engineer to Senior Systems Engineer to Technology Analyst, on the strength of manager feedback on delivery quality.
- Rewrote XQuery transformation logic, with supporting Java and SQL development, across 30+ live insurance document templates for Aetna's platform on Majesco Cloud Insurer, cutting document generation time from 5 minutes to 1 minute and reducing recurring formatting defects by 80%.
- Validated and troubleshot 25+ monthly production releases using SVN, Jenkins and Majesco DaaS deployment tooling, resolving billing discrepancies by pulling records via Microsoft SQL Server and pushing business-approved corrections through an existing REST API, and confirming each release's changes reached Aetna's live platform as intended.
- Partnered with business analysts, QA engineers, frontend developers, and integration teams on a 20-person Agile team, serving as one of two dedicated engineers on the insurance document-generation (DaaS) platform.
- Delivered feature and enhancement work across agent onboarding, group setup and enrolment, and claims-upgrade merges, translating shifting regulatory and business requirements into production code.
- Mentored 3 junior engineers on XQuery development and DaaS system architecture, onboarding them to the codebase and reviewing their work, cutting their ramp-up time from roughly 4 weeks to 2 weeks.

Team Member (Part-time) | September 2025 – Present

Oak Fire Pizza, Cork, Ireland

- Developed strong teamwork, communication, time-management, and multitasking skills while working effectively under pressure in a fast-paced environment.

Technical Skills

Languages: Java, Python, SQL, XQuery, TypeScript, C, C++, HTML, CSS
Frameworks: Angular, Django, Spring Boot, PyTorch, Bootstrap, PySpark
Developer Tools: Git/GitHub, RESTful API, Postman, VS Code, NetBeans, Altova XMLSpy, Google Colab, Ollama, Microsoft SQL Server, MySQL, SVN, Jenkins, Claude Code, Codex
Libraries: Hugging Face Transformers, scikit-learn, pandas, NumPy, Matplotlib, Captum, OpenAI Whisper

Projects

Multimodal Depression and Positive Affect Detection | *PyTorch, Whisper, Captum, Transformers, Scikit-learn*

- Designed and built a multimodal transformer system for depression and positive-affect detection, fusing a fine-tuned XLM-RoBERTa text encoder, a residual audio-visual fusion network, and a three-way MentalRoBERTa ensemble, trained across 909 Reddit users, 4,999 mental-health posts, and 773 YouTube vlog transcripts.
- Addressed a severe 6% positive-affect class imbalance by combining cross-entropy, LLM-augmented, and class-balanced training into a three-way MentalRoBERTa ensemble, using a locally-run Mistral model via Ollama to generate synthetic training examples.
- Fused the three frozen encoders through learned projection heads and a shared classifier into a single multimodal backbone, reaching AUROC 0.987 and F1 0.951, a 23.5-point AUROC gain over the strongest traditional baseline.
- Validated model decisions with Integrated Gradients attribution across all three datasets, showing recovery-oriented language (progress, hope, courage) consistently increased positive-affect scores and distress language (afraid, anxious, tired) consistently decreased them, in both Reddit text and Whisper-transcribed vlog speech.

Stock Market Analysis and Price Prediction | *Python, Django, Machine Learning, Web Scraping*

- Built a Django web application ingesting 5 years of daily OHLCV data via yfinance for any NSE-listed ticker on demand, with screener and news features covering the fixed NIFTY 50 index (50 stocks), deployed live on an Oracle Cloud VM with HTTPS.
- Trained a stacked LSTM network (128 to 64 units, followed by two dense layers) on historical price data to generate stock-price predictions, evaluating each prediction's accuracy with RMSE computed live against a held-out test split.

Parking Management System | *Java, MySQL, Spring Boot*

- Built a full-stack Spring Boot parking management system on a 9-table MySQL schema (users, customers, vehicles, vehicle brands/models, rates, slots, bookings, payments), deployed live on an Oracle Cloud VM with HTTPS.
- Implemented session-based authentication through a custom interceptor with SHA-256 password hashing, controlling access across booking and payment operations.
- Built a real-time operations dashboard showing available and occupied slot counts, active bookings, and daily and total revenue, alongside a booking-history view for past transactions.

Certifications

Infosys Certified Database and SQL Professional (July 2024)
Infosys Certified Python Associate (May 2024)
Infosys Certified Agile Developer (April 2022)